Subgroup Performance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach subgroup comparisons with all needed records.

Supply complete metrics and source IDs,explain small/empty groups and keep exploratory overlap separate from pooled evidence.

Keep one habit: Require denominators,label quality,context and bounded conclusions before a release discussion.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Calculate **subgroup** results with sample sizes

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **subgroup** evaluation. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Different error patterns in different evaluated conditions or groups.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Opening retrieval

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Look for: Plan one **evidence**-led change and test it on source-separated validation cases

Reference example, not the only acceptable wording: Name the exact hypothesis,changed factor and fixed comparison protocol. In the synthetic crossed-background trial,**training** additions change the chosen feature and improve four validation views from two plants;this does not establish real diagnosis or final reliability. Preserve all outputs,source groups and fresh final-evaluation needs.

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Look for: Read **training**-size learning curves

Reference example, not the only acceptable wording: Teacher explanation: Mean(5+7+6)/3=6; range 7-5=2.

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a declared score threshold including equality

Reference example, not the only acceptable wording: Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

End goal

By the end, I can explain **subgroup** performance using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Improve Generalization

Learning Curves

Metrics Match Consequences

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

Concept to explain

Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

Term	Meaning
subgroup	meaningful slice of evaluation cases
performance gap	difference between subgroup metrics
denominator	number of cases behind a rate
intersection	cases belonging to multiple slices

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach weighted pooling and denominator sensitivity first

> Keep per-case source IDs,positive counts and correlation/causation distinctions visible.

Main object-condition scope and complete aggregate table

A single overall score can hide different experiences. Define relevant groups before evaluation and report counts as well as rates. This classroom exercise uses fictional sensor conditions, not personal identities.

Real evaluations involving people require appropriate permissions, privacy and contextual expertise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **subgroup** evaluation.

Main sensor/camera slice	Observed errors	Evaluated count	Error rate / accuracy
A	2	20	10% /90%
B	3	5	60% /40%
Pooled disjoint A/Bcases	5	25	20% /80%
Observed error-rate difference	Bminus A	Same error definition	50 percentage points

Sensor group A has 2 errors in 20 examples and group B has 3 in 5. Error rates are 10 and 60 percent; pooled error is 5/25=20 percent. The group difference is 50 percentage points.

B has only five examples, so its rate is sensitive to each individual result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Term	Meaning
Subgroup	A defined subset of the evaluated examples
Disaggregation	Reporting results separately for defined groups
Coverage	Which groups and conditions are represented
Disparity	An observed difference between group results

Subgroup evaluation at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Group A	2 errors of 20	10 percent
Group B	3 errors of 5	60 percent
Pooled	5 errors of 25	20 percent

Changing one B error to a correct result changes its rate from 60 to 40 percent. Small samples can be unstable without making the observed errors unimportant. A missing group has no measured rate, not zero error. Intersections of conditions may be even smaller.

How does correcting one B error affect its rate? Explain the steps or evidence behind your response.

Read pictured added-case comparison

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

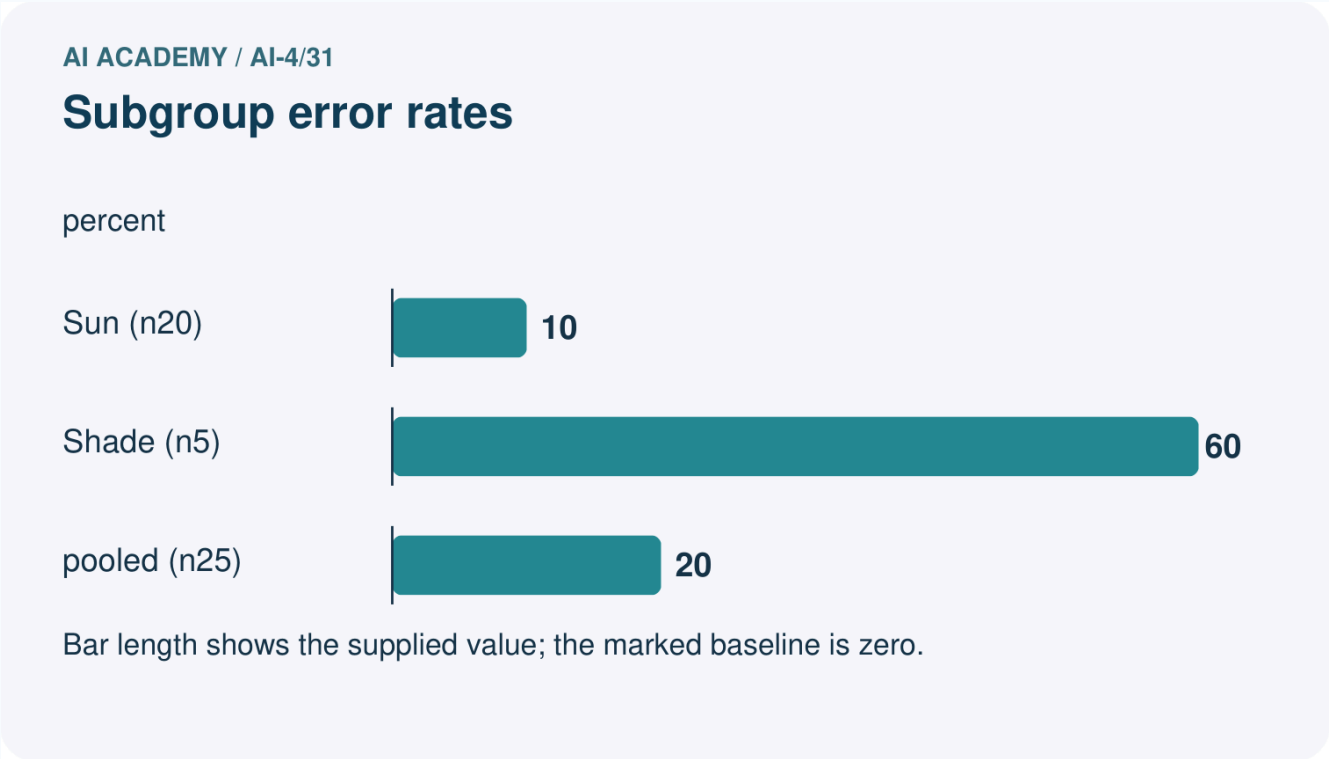
Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Picture Sun2/20 errors and Shade3/5 match the main 10%/60%arithmetic; pooled 5/25=20%, gap 50 percentage points. Adding five new Shadecases with no errors changes Shade to3/10=30%and pooled to5/30≈16.7%, without removing any original error. The model was

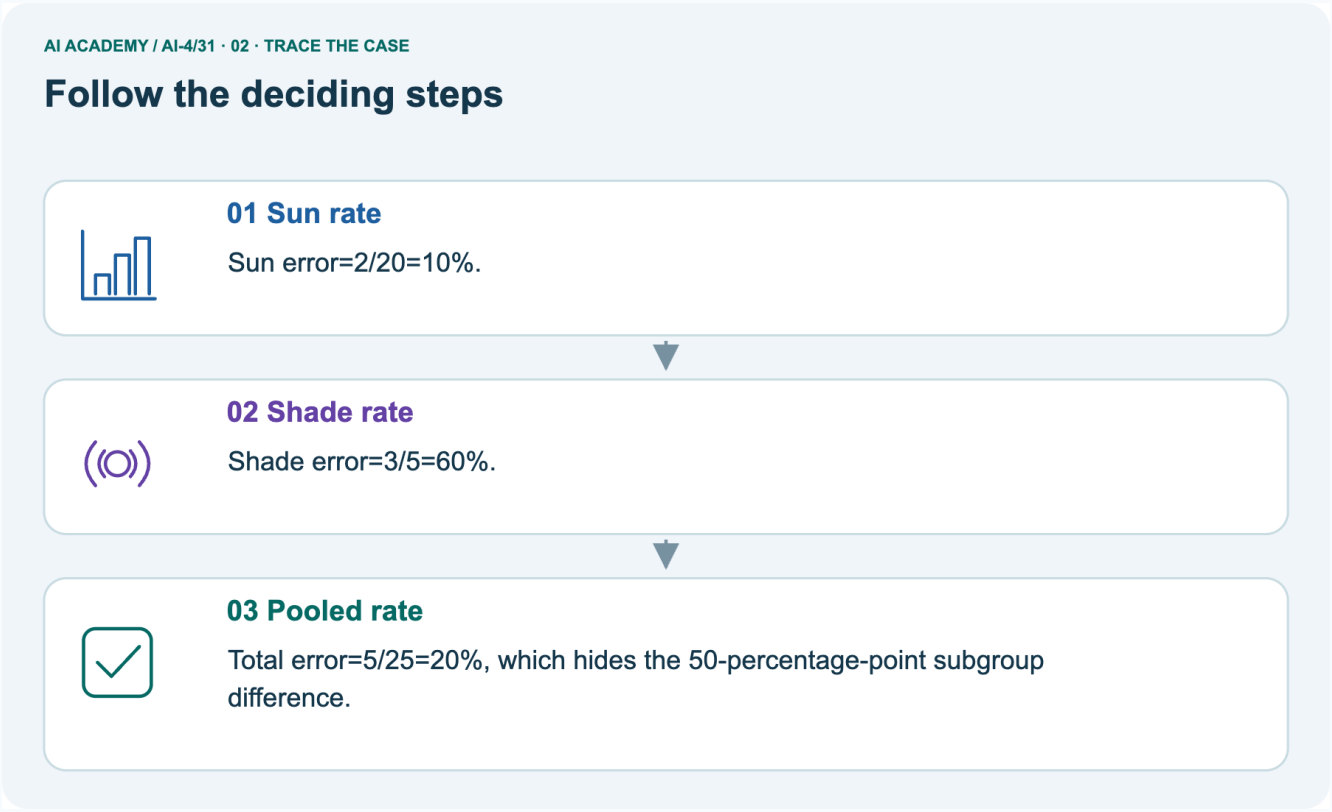
not improved merely by changing the evaluation sample. New per-class labels are not supplied for those additions, so do not invent revised recall.

Correcting one of the original five Shadeoutputs instead would give 2/5=40%; record a reviewed output or model-version change separately.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace subgroup rates and unchanged errors



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Sun rate	Sun error=2/20=10%.
2. Shade rate	Shade error=3/5=60%.
3. Pooled rate	Total error=5/25=20%, which hides the 50-percentage-point subgroup difference.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Small **subgroup** samples have high variability. An observed gap does not by itself establish discrimination or a causal explanation.

Change: Shade gains5 new cases with no errors.

Prediction: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.

Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

▼ Additional18/20versus6/10accuracy example

A has18/20 correct; B has6/10.

Compute **subgroup** results and **evidence** limits.

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Calculate **subgroup** accuracy and recall gaps with adequate **denominator**s.

MODEL IDEA Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Daylight: 18/20 correct = 90%; evening: 10/20 = 50%. Accuracy gap = 90% – 50% = 40 percentage points.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Daylight: 18/20 correct = 90%; evening: 10/20 = 50%.

STEP 2 Accuracy gap = 90% – 50% = 40 percentage points.

STEP 3 Evening spotted recall is 4/10 = 40%, versus daylight 8/10 = **80%**.

STEP 4 Inspect representation and image quality; report counts and uncertainty before repairing.

CHECK Verify the **denominator**, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

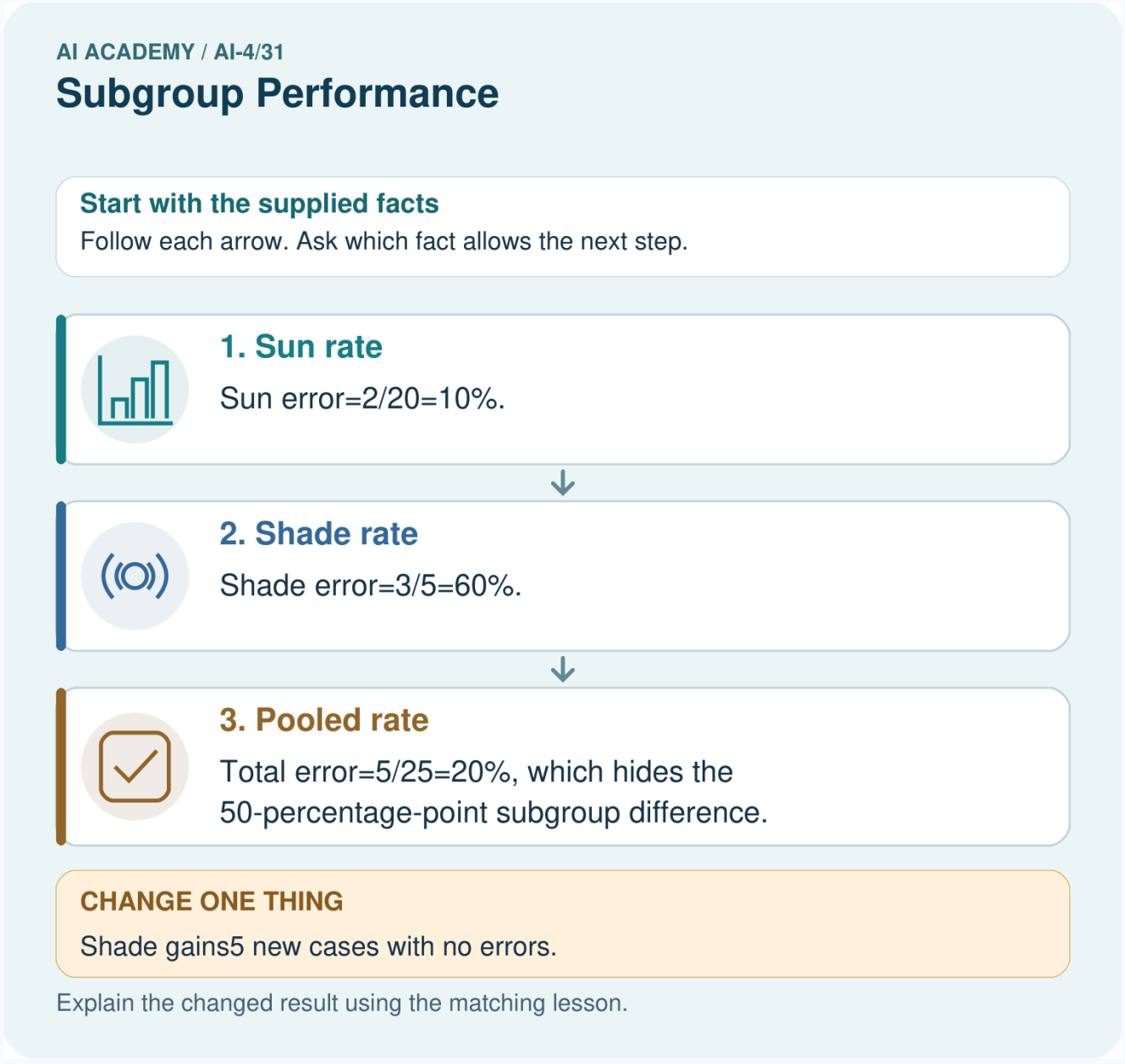
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Subgroup Performance

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: How does correcting one B error affect its rate?

Teacher check: Teacher explanation: B becomes 2/5=40 percent, a 20-percentage-point change from one case.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Subgroup** evaluation. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The20% original total is dominated by Sun's larger sample and understates the observed Shade difficulty. Include counts, conditions and uncertainty.

Changed-case check: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Changed-case reasoning: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: A has18/20 correct; B has6/10.

Write the question to explain: Compute **subgroup** results and **evidence** limits.

Reveal after **prediction**s: 90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

Use the earlier transfer question as additional practice. Reserve the selected fresh X1/10andY2/4error-rate task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and **denominator**s for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: B becomes 2/5=40 percent, a 20-percentage-point change from one case. Exit response: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: What can a pooled error rate hide? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify subgroup in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: calculate group rates; check sensitivity. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between subgroup and disaggregation. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Group C has no evaluated cases. What is its measured error rate?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Calculate subgroup results with sample sizes; Distinguish missing evidence from equal performance. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Full audit counts as supplied; cameraChas no evidence, not perfect performance. Alpha/Betaequal20%error but different 100%/75%recalls, showing metrics answer distinct questions. Prioritize a source-aware investigation, not an inherent group cause. IndependentX10%, Y50%, pooled 3/14≈21.4%; Yonly4 cases. No class counts for recall and no causal explanation are supplied.

Keep predeclared versus exploratory slices explicit.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. Slice Audit: compute metrics and **denominator**s for lighting, species, and camera; flag small samples and prioritize one gap.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

A stronger claim makes a stronger project: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent.

Y has only four examples, and the table does not establish a cause.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh X1/10andY2/4error-rate task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A good overall result guarantees acceptable performance for every **subgroup**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

Teacher explanation: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent. Y has only four examples, and the table does not establish a cause.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Calculate group rates

Calculate the worked group and pooled rates.

Teacher explanation: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

W2 Check sensitivity

How does correcting one B error affect its rate?

Teacher explanation: B becomes 2/5=40 percent, a 20-percentage-point change from one case.

W3 Report missing coverage

Group C has no evaluated cases. What is its measured error rate?

Teacher explanation: Unavailable because the **denominator** is zero; report missing **coverage** and do not substitute zero.

W5 Avoid a fairness shortcut

Why does equal accuracy not settle whether use is fair?

Teacher explanation: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

Additional transfer practice question

Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

Reasoned practice answer

Bright: 90%; dim: **60%**; gap: 30 percentage points. Report **denominator**s and uncertainty from small samples before making broad claims.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The difference is 30 percent of all examples rather than 30 percentage points between rates.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Overall=24/30=**80%**; A=90%, B=**60%**, gap=30 percentage points. These data do not prove the device itself causes the difference; lighting, leaf mix or labels may differ.

Investigate with comparable cases and preserve small-sample uncertainty.

Explanations for the additional workbook examples

W1 Calculate group rates

W2 Check sensitivity

W3 Report missing coverage

W4 Apply independently

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

Teacher explanation: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent. Y has only four examples, and the table does not establish a cause.

W5 Avoid a fairness shortcut

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Add **subgroup** tables and a minimum acceptable threshold to the Plant Classifier release criteria.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and **subgroup** or shifted-condition analysis with **denominator**s.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **subgroup** and **disaggregation** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent. Y has only four examples, and the table does not establish a cause.

Fresh retry X 2/8 and Y 1/2:25 and 50 percent, pooled 3/10=30 percent. Require **denominator**s, the small Y sample and no causal inference. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/32, ai-4/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 32: Proxy Features and Fairness. Retrieve today's **evidence** and identify what carries forward and what is new.

Next week: Proxy features and fairness. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Calculate **subgroup** results with sample sizes

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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Before class, solve the supported practice: How does correcting one B error affect its rate?

Reasoned practice answer: Teacher explanation: B becomes 2/5=40 percent, a 20-percentage-point change from one case.

Check every learner before discussion: Use the worked case for **Subgroup** evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Shade gains5 new cases with no errors. Require a **prediction** and a reason before comparing. Expected result: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.

Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

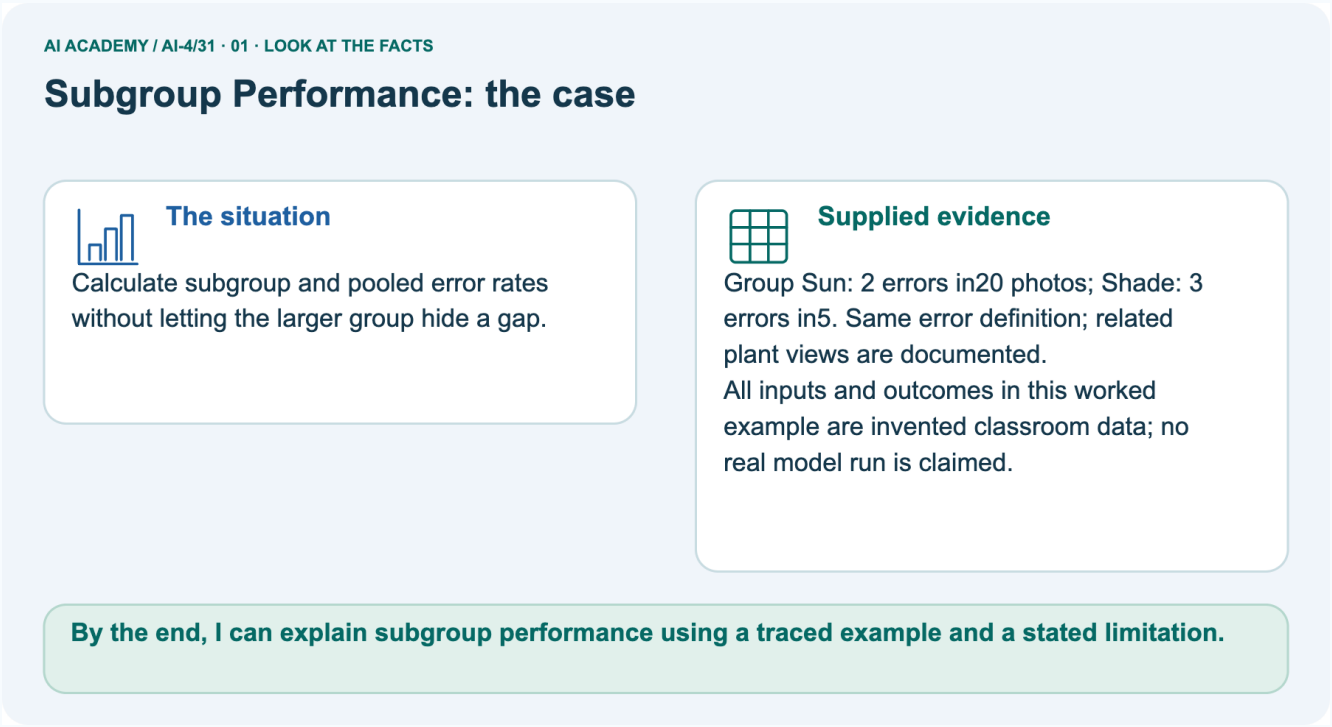
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Subgroup Performance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

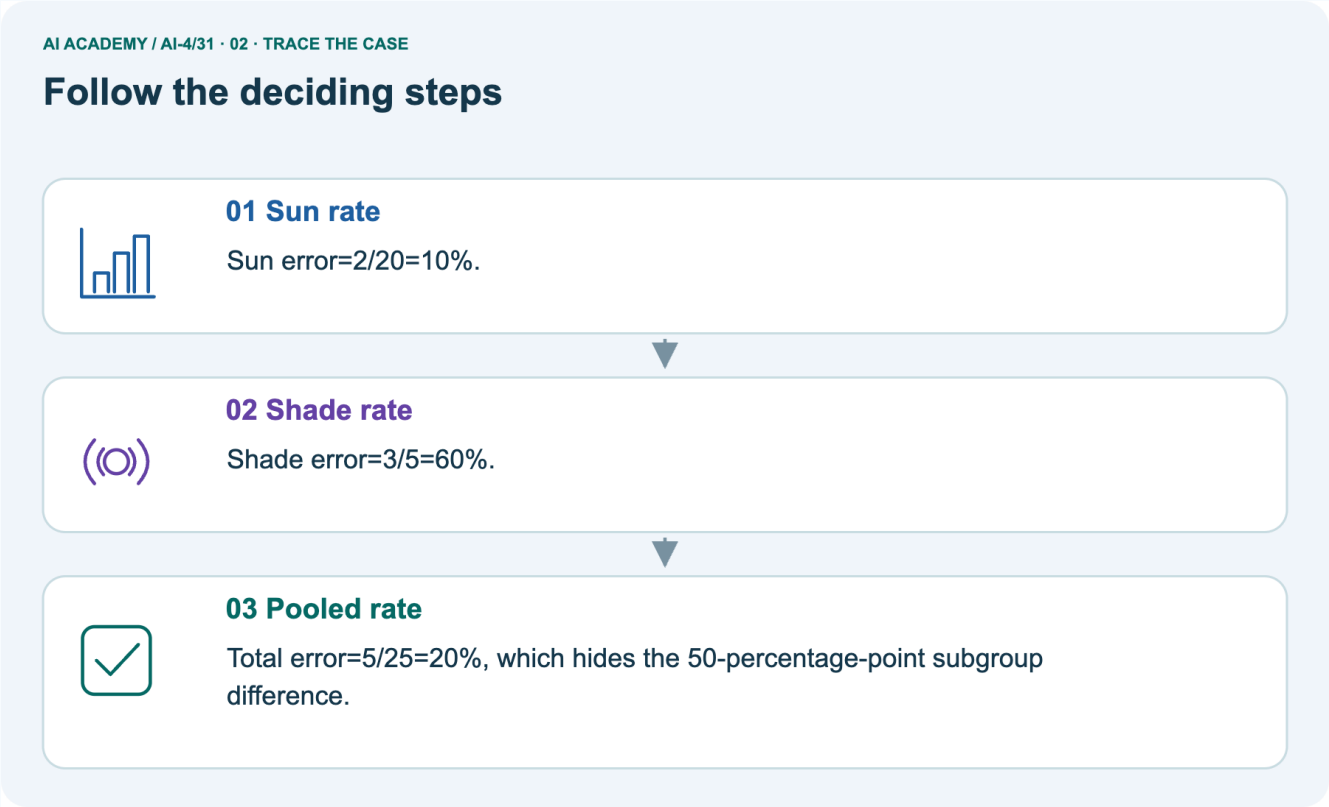
Calculate **subgroup** and pooled error rates without letting the larger group hide a gap. Supplied facts: Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Sun rate: Sun error=2/20=10%. → Shade rate: Shade error=3/5=**60%**. → Pooled rate: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Subgroup error rates



Follow the labelled picture with the matching explanation. Open the vector diagram

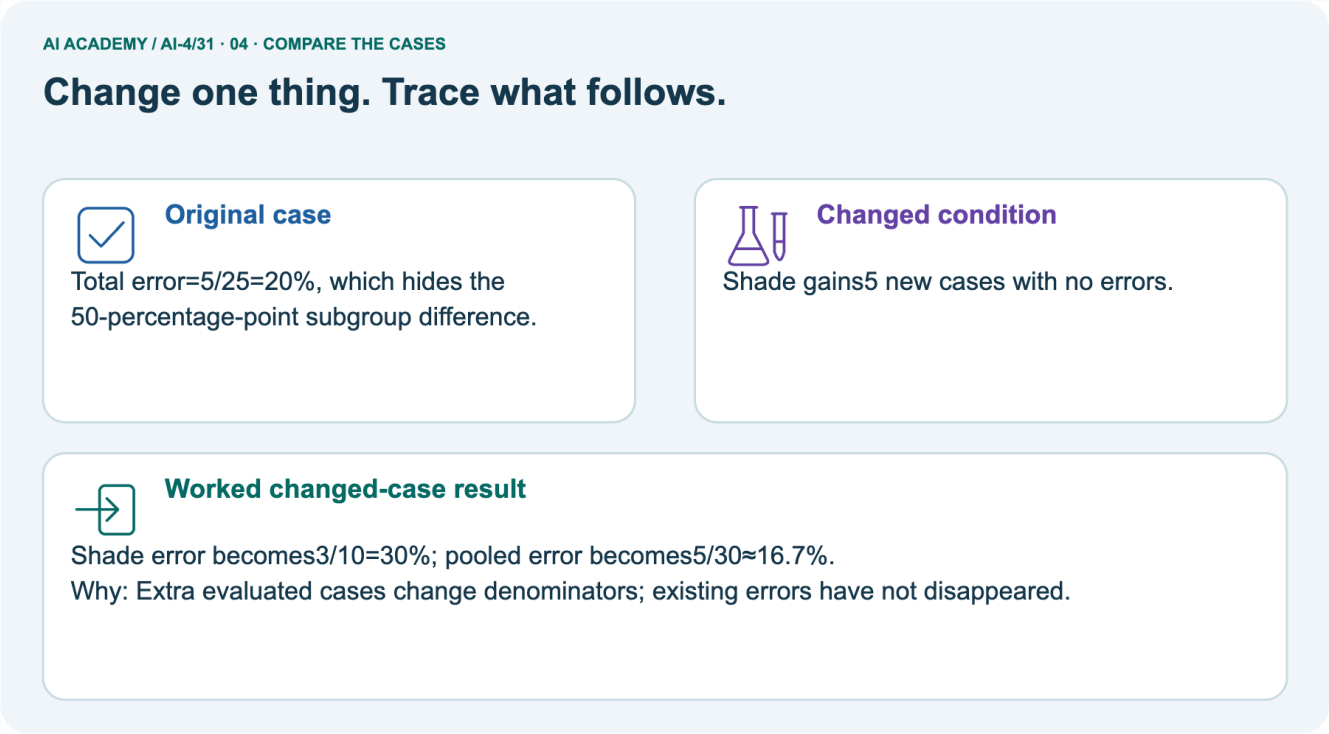
Subgroup error rates. Units: percent. Sun (n20): 10. Shade (n5): 60. pooled (n25): 20.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and **denominator**s before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Changed condition: Shade gains5 new cases with no errors.

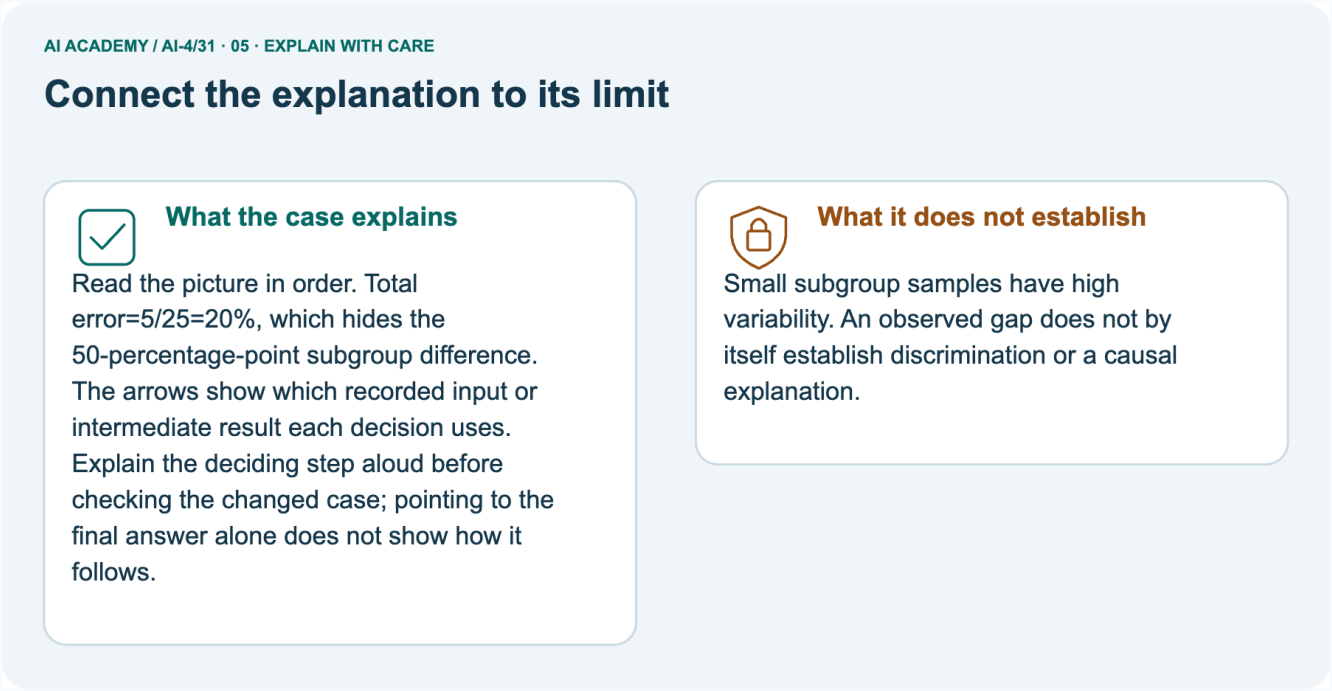
Expected result: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Small **subgroup** samples have high variability.

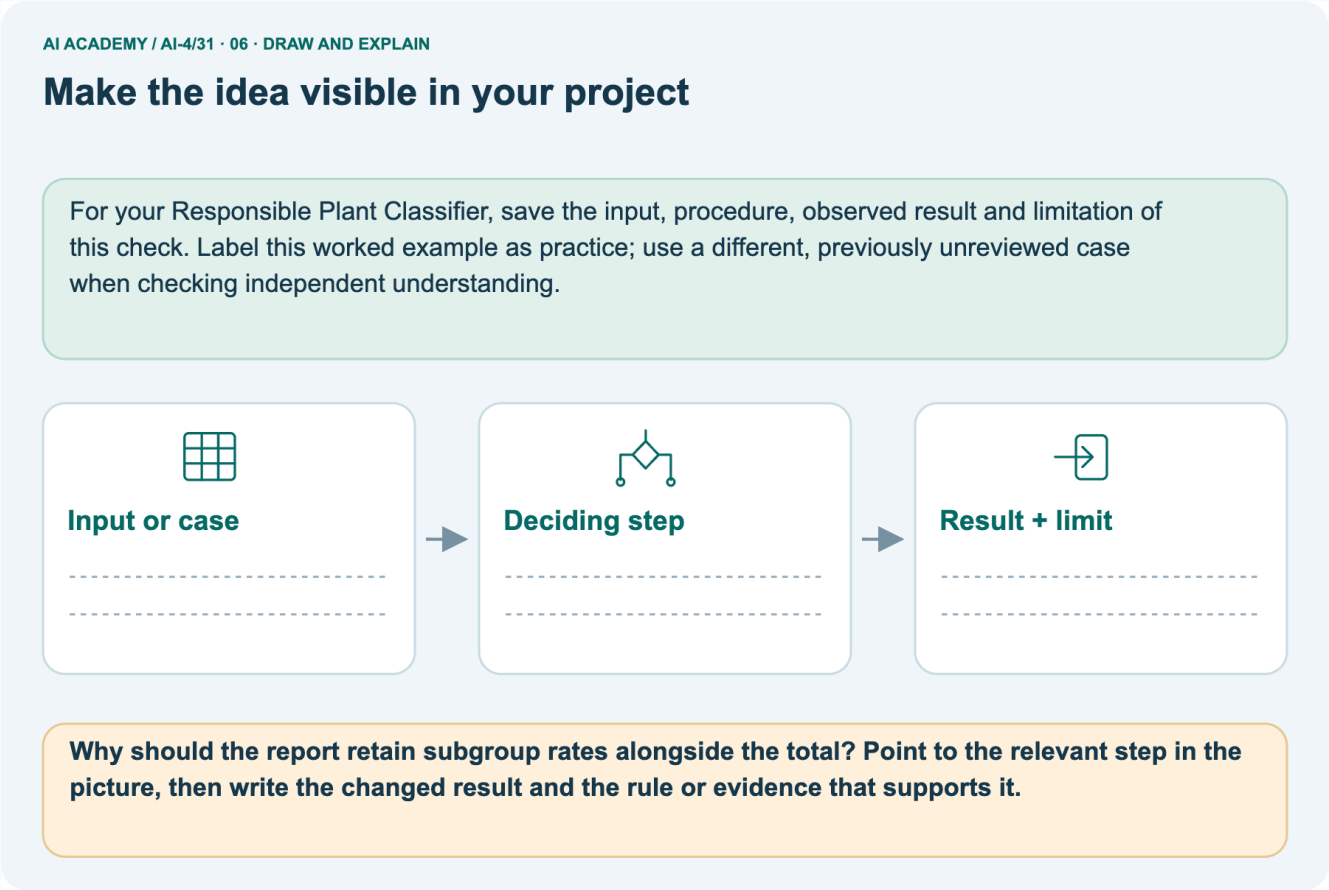
An observed gap does not by itself establish discrimination or a causal explanation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Small **subgroup** samples have high variability. An observed gap does not by itself establish discrimination or a causal explanation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The20% original total is dominated by Sun’s larger sample and understates the observed Shade difficulty. Include counts, conditions and uncertainty.

Changed-case check: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Subgroup Performance: the case

02. Follow the deciding steps

03. Subgroup error rates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.